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H. 220 Trees Reassessed

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with ***N*** nodes. You are given the in-order and post-order traversals of the tree. Your task is to determine the pre-order traversal of the tree.

Input

The first line contains an integer N $(1 \leq N \leq 1000)$ — the number of nodes in the binary tree.

In the next line, there will be N distinct integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq N)$ separated by spaces – representing the in-order traversal of the tree.

The following line, there will be N distinct integers $b_1, b_2, b_3 \ldots b_n$ $(1 \leq b_i \leq N)$ separated by spaces – representing the post-order traversal of the tree.

Output

Print N space-separated integers representing the pre-order traversal of the binary tree.

Examples

input	Copy
5 1 2 3 4 5 1 4 5 3 2	

output	Copy
2 1 3 5 4	

input	Copy
10 1 2 3 4 5 6 7 8 9 10 2 1 3 5 6 7 4 10 9 8	

output	Copy
8 4 3 1 2 7 6 5 9 10	

CSE221 Assignment 03 Fall 2026
Contest is running
6 days
Contestant

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Language: Java 21 64bit
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